St. George's University
Grenada, West Indies

Basic Principles of Medicine 2

DM: Digestive System & Metabolism | Lecture 3

Anatomy of the Stomach, Liver and Gallbladder

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Your Objectives will show here!

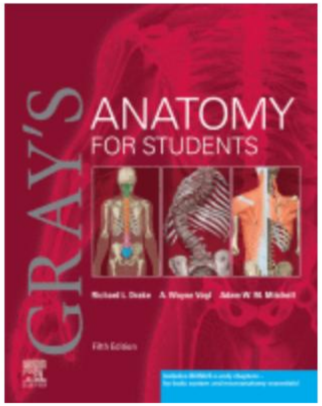
Objectives

Foregut Structures

SOM.MK.1.BMP2.3.DM.1.ANAT.1125	Describe the following ligaments, their embryonic origin, peritoneal coverings and their function: falciform, round ligament of the liver (ligamentum teres hepatis), coronary, triangular, median, medial and lateral umbilical, greater omentum, gastrocolic, gastrosplenic, splenorenal ligaments, lesser omentum (hepatoduodenal and hepatogastric ligaments) and suspensory ligament of the duodenum (Treitz).
SOM.MK.1.BMP2.3.DM.1.ANAT.1126	Describe the abdominal esophagus, its blood supply and lymphatic drainage.
SOM.MK.1.BMP2.3.DM.1.ANAT.1127	Describe the nerve pathways for pain in a patient with gastroesophageal reflux disease.
SOM.MK.1.BMP2.3.DM.1.ANAT.1129	Describe sliding and paraesophageal hiatal hernia
SOM.MK.1.BMP2.3.DM.1.ANAT.1130	Describe the anatomy of the stomach and identify its parts and curvatures.
SOM.MK.1.BMP2.3.DM.1.ANAT.1131	Describe the relationships of the stomach to the spleen, pancreas, liver, ascending and descending colon.
SOM.MK.1.BMP2.3.DM.1.ANAT.1132	Describe the blood supply and the venous and lymphatic drainage of the stomach.
SOM.MK.1.BMP2.3.DM.1.ANAT.1159	Describe the location of the liver and its relationship to surrounding structures.
SOM.MK.1.BMP2.3.DM.1.ANAT.1160	Describe the four anatomical lobes of the liver.
SOM.MK.1.BMP2.3.DM.1.ANAT.1161	Describe the functional (physiological) lobes of the liver.
SOM.MK.1.BMP2.3.DM.1.ANAT.1162	List the structures entering / leaving the liver at the porta hepatis.
SOM.MK.1.BMP2.3.DM.1.ANAT.1163	Describe the contents of the hepatoduodenal ligament which will be gripped during a Pringle maneuver.
SOM.MK.1.BMP2.3.DM.1.ANAT.1164	Describe the "the bare area" of the liver.
SOM.MK.1.BMP2.3.DM.1.ANAT.1165	Describe the blood supply, venous and lymphatic drainage to the liver.
SOM.MK.1.BMP2.3.DM.1.ANAT.1166	Describe the contents of a liver segment and its importance in surgery of the liver.
SOM.MK.1.BMP2.3.DM.1.ANAT.1167	Describe the location of the gall bladder and its relationship to the liver.
SOM.MK.1.BMP2.3.DM.1.ANAT.1168	Describe the anatomy of the biliary tree.
SOM.MK.1.BMP2.3.DM.1.ANAT.1169	Describe the arterial supply, venous and lymphatic drainage of the gall bladder and cystic duct.
SOM.MK.1.BMP2.3.DM.1.ANAT.1172	List the borders of the triangle of Calot and state the importance of this triangle when performing laparoscopic surgery for gallbladder disease

Recommended Reading

- Drake, Vogl, Mitchell. Grays Anatomy for Students. 5th ed. Elsevier Churchill Livingstone.
- Online Textbook: Chapter 4: Click here
  - Abdominal Viscera
    - Abdominal Organs
      - Esophagus, Stomach
      - Liver, Gallbladder



- Pay special attention to the green (print text)/blue (ebook)"In the Clinic" boxes for the clinical correlations for this topic

Foregut

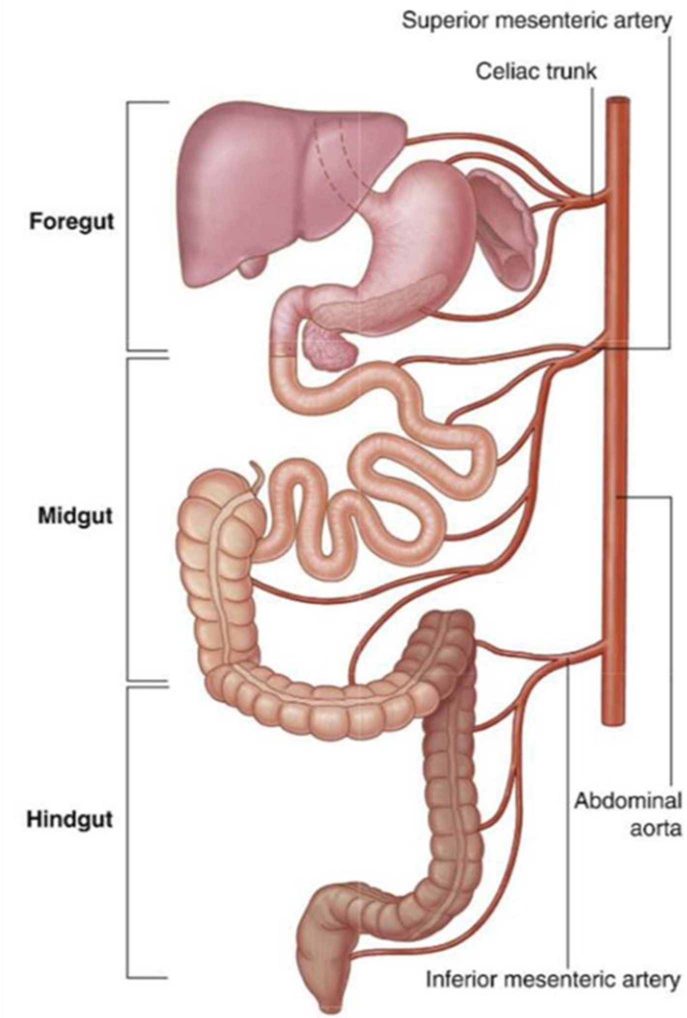
Begins at the abdominal esophagus and ends proximal to the major duodenal papilla.

- Esophagus
- Stomach
- Liver /Biliary apparatus
- Pancreas
- Spleen (developed in dorsal mesogastrium)
- ½ Duodenum (proximal to the major papilla)

Arterial supply: Celiac Trunk

Venous drainage: Portal Venous System

Lymphatic drainage: Celiac Nodes



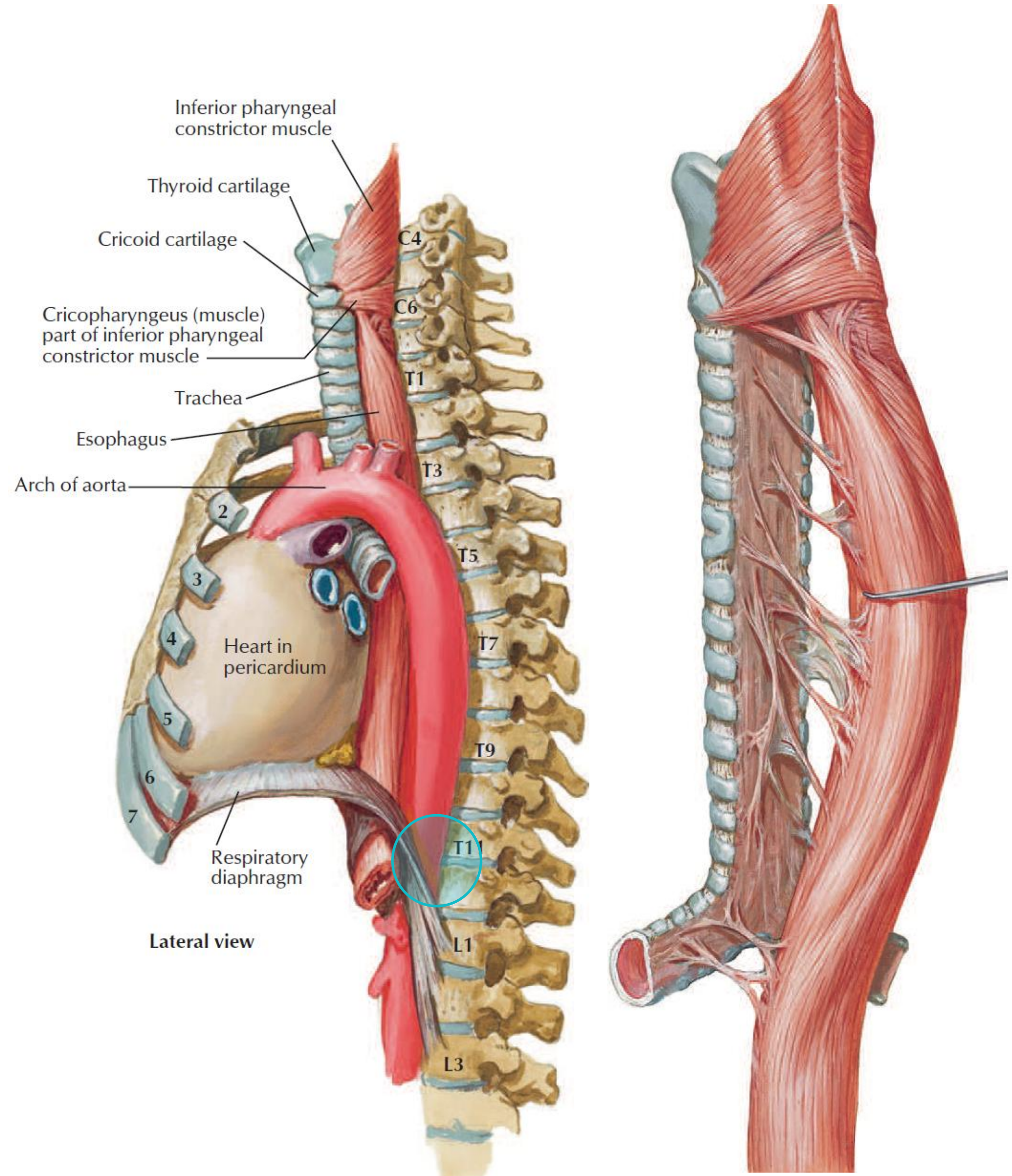
Esophagus

- Part of foregut
- 25cm long
- Location-neck, thorax & abdomen

Musculature of esophagus :

- Skeletal – upper 1/3
- Mixed – middle 1/3
- Smooth – lower 1/3

- Pierces the diaphragm at the level of T10
- Below the diaphragm, is covered by peritoneum.



Esophageal Constrictions

1. Upper Esophageal Sphincter
2. Arch of the Aorta
3. Left Main Bronchus
4. Lower Esophageal Sphincter

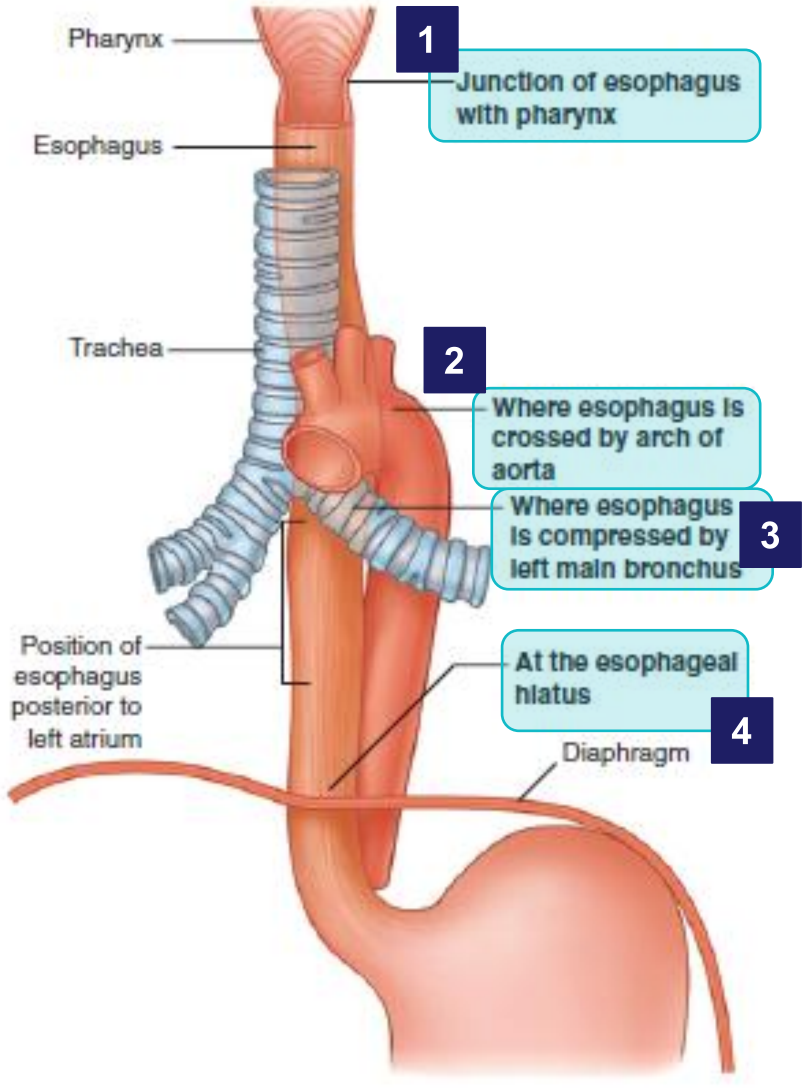
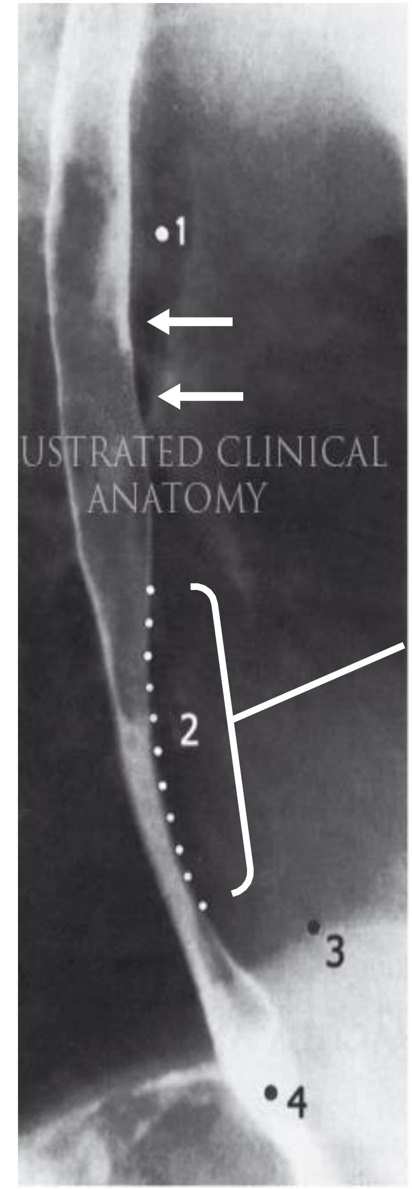
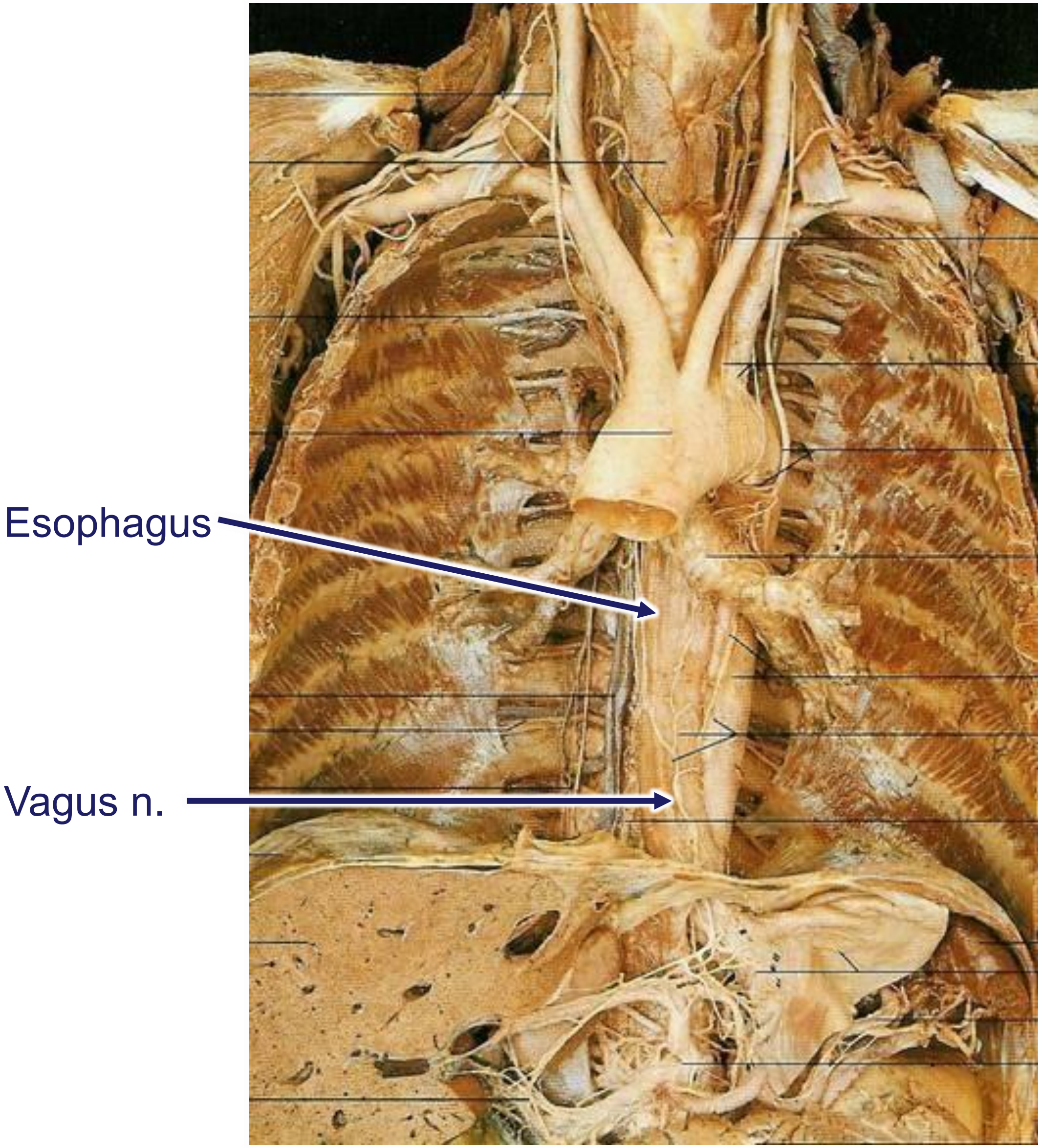
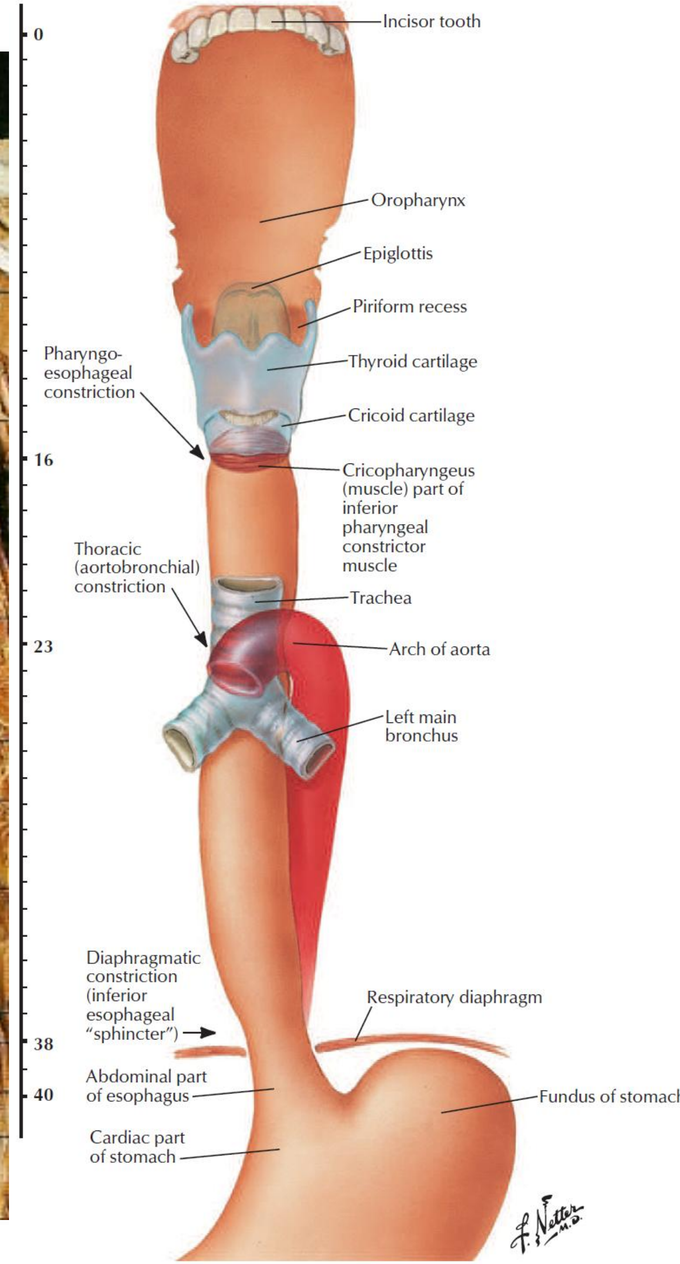


Fig. 3.98 Sites of normal esophageal constrictions.



Esophagus


Esophagus
Vagus n.


0
16
23
38
40
Incisor tooth
Oropharynx
Epiglottis
Piriform recess
Thyroid cartilage
Cricoid cartilage
Pharyngo-esophageal constriction
Cricopharyngeus (muscle) part of inferior pharyngeal constrictor muscle
Thoracic (aortobronchial) constriction
Trachea
Arch of aorta
Left main bronchus
Diaphragmatic constriction (inferior esophageal "sphincter")
Respiratory diaphragm
Abdominal part of esophagus
Fundus of stomach
Cardiac part of stomach

Esophageal Supply

Blood Supply

- Esophageal arteries (thoracic aorta, L. gastric a.)
- Esophageal veins (drain into left gastric vein)

Lymphatic Drainage

- Posterior mediastinal + Left gastric nodes

Nervous Supply

- Esophageal complex (esophageal branches from vagal trunks + sympathetic trunks)
- Visceral afferents (vagus, sympathetics, splanchnic nerves)

Pain in GERD

- Visceral afferents that pass through **sympathetic trunks**

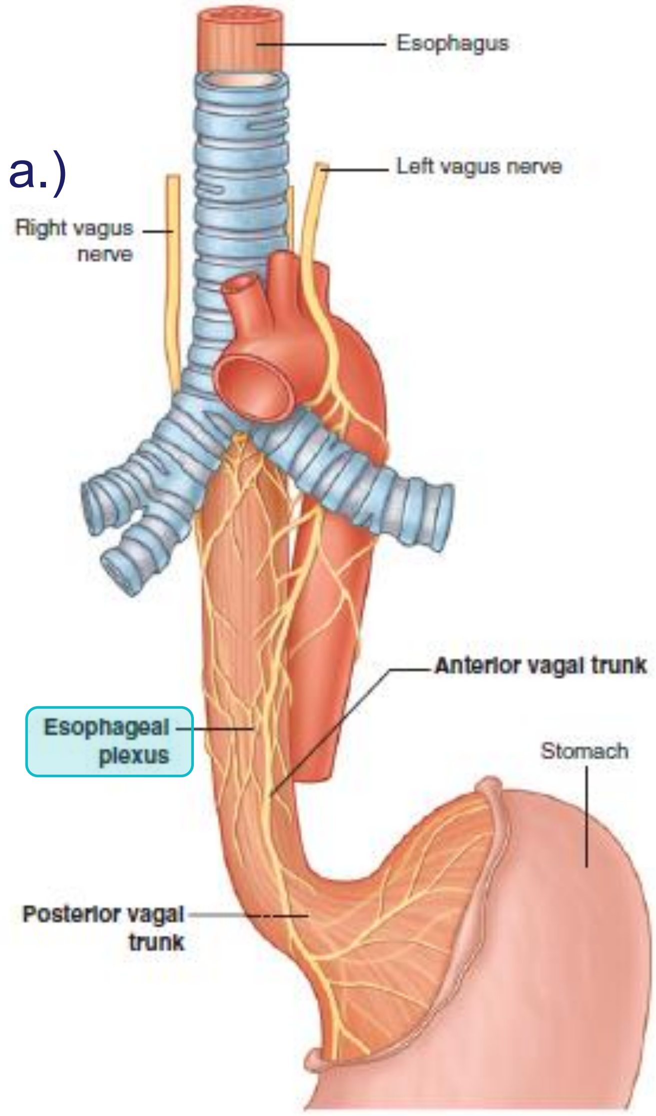
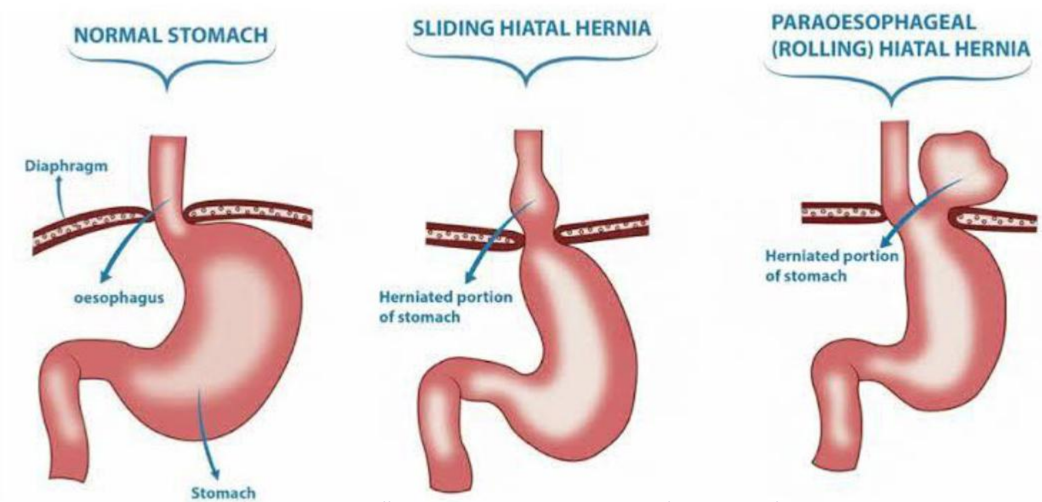


Fig. 3.99 Esophageal plexus.

Esophagus – Hiatal Hernia

- At the level of the esophageal hiatus
- Lax in diaphragm → fundus (stomach) herniates into posterior mediastinum
- **Sliding hernia (more common)**:
  - Esophagus and stomach slide up into chest
  - Gastroesophageal junction displaced upward
  - Associated with GERD
- **Paraesophageal hernia**
  - Gastroesophageal junction remains in normal position
  - Fundus of stomach slides up into chest next to (-*para*) esophagus



https://drvasudevgastroenterologist.com/hiatal-hernia/

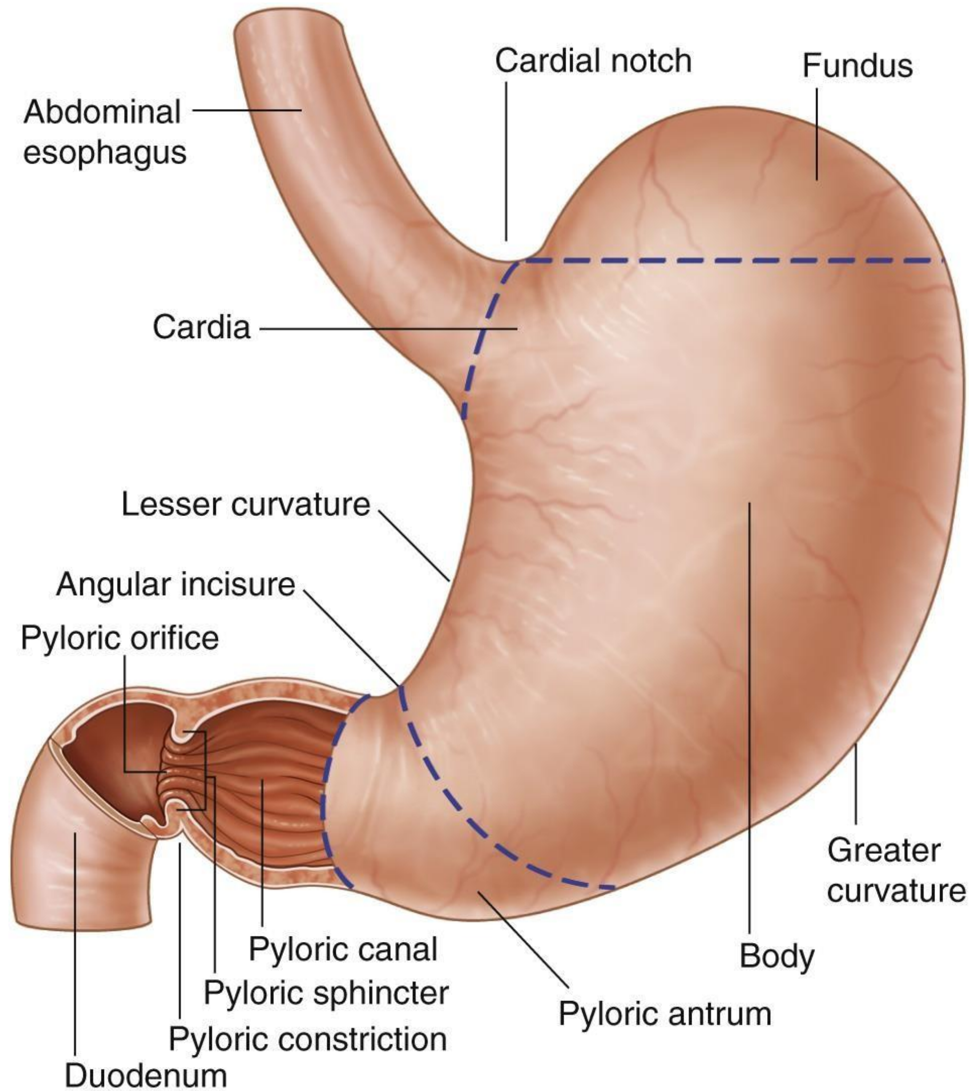
Stomach

Parts

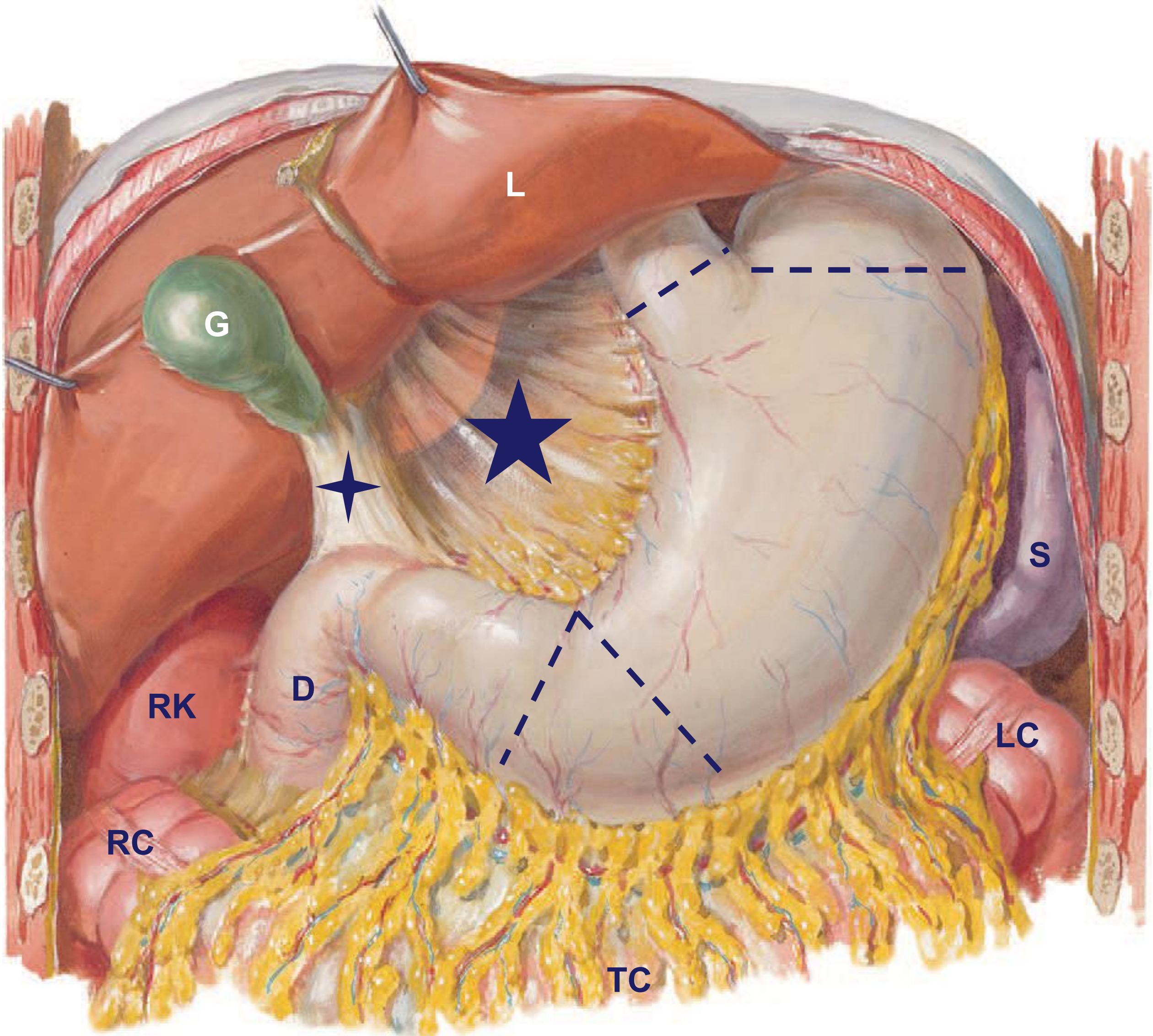
- Cardia
- Fundus
- Body
- Pyloric region
    - Antrum
    - Canal
    - Sphincter

Curvatures

- Lesser curvature
- Greater curvature



Stomach Relationships



Lesser Omentum

- Hepatogastric ligament ★
- Hepatoduodenal ligament ✦

G – Gallbladder

L – Liver

RC – Right colic flexure

RK – Right kidney

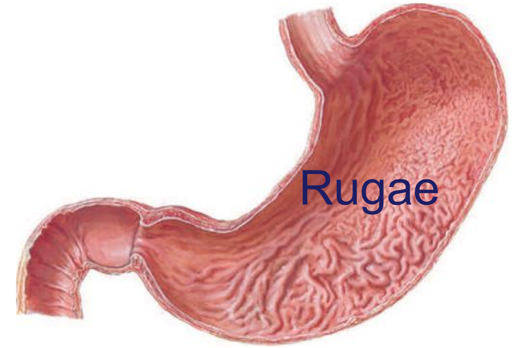
D – Duodenum

TC – Transverse colon (behind to greater omentum)

S – Spleen

LC – Left colic flexure

Stomach – Internal View

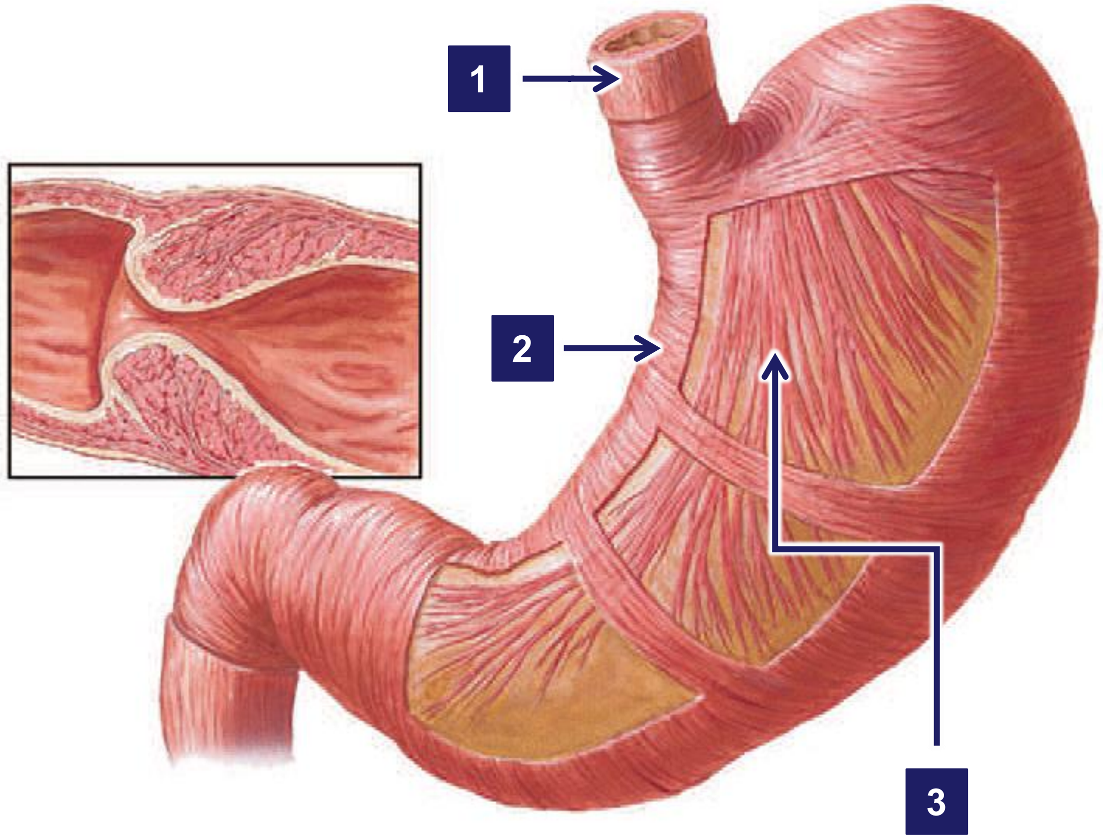


Musculature

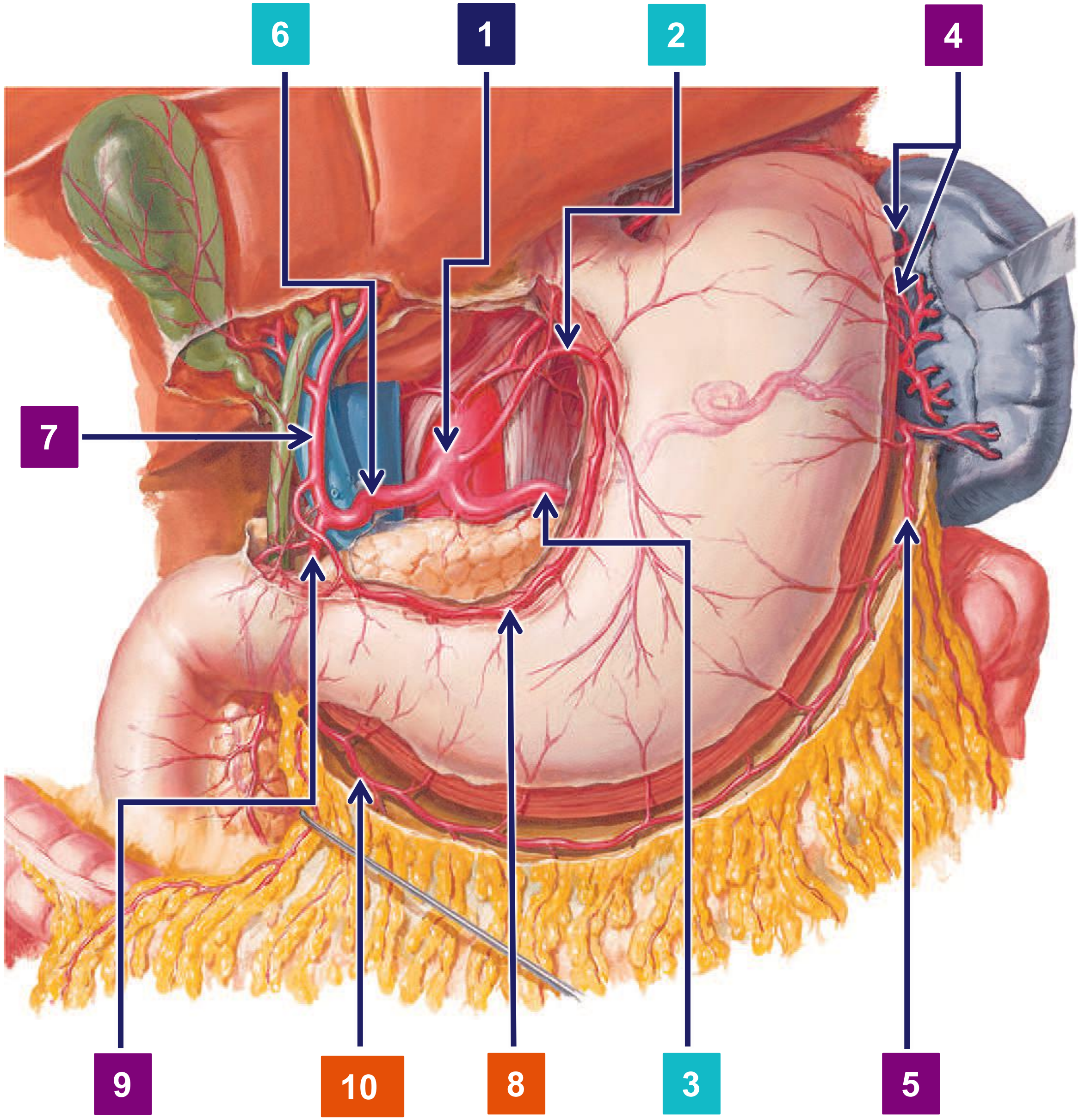
1. Outer longitudinal
2. Middle circular
3. Innermost oblique

Pyloric Sphincter

- Thickened ring of circular muscle



Stomach – Arterial Supply

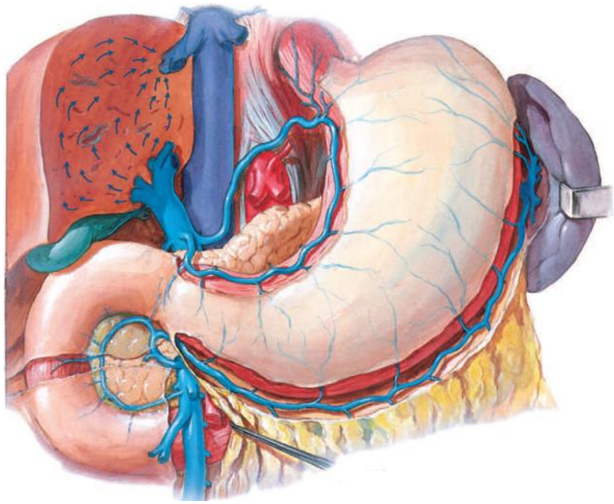


1. **Celiac trunk**
2. Left gastric
3. Splenic
   4. Short gastrics
   5. L. gastro-omental
6. Common hepatic
   7. Proper hepatic
      8. Right gastric
   9. Gastroduodenal
      10. R. gastro-omental

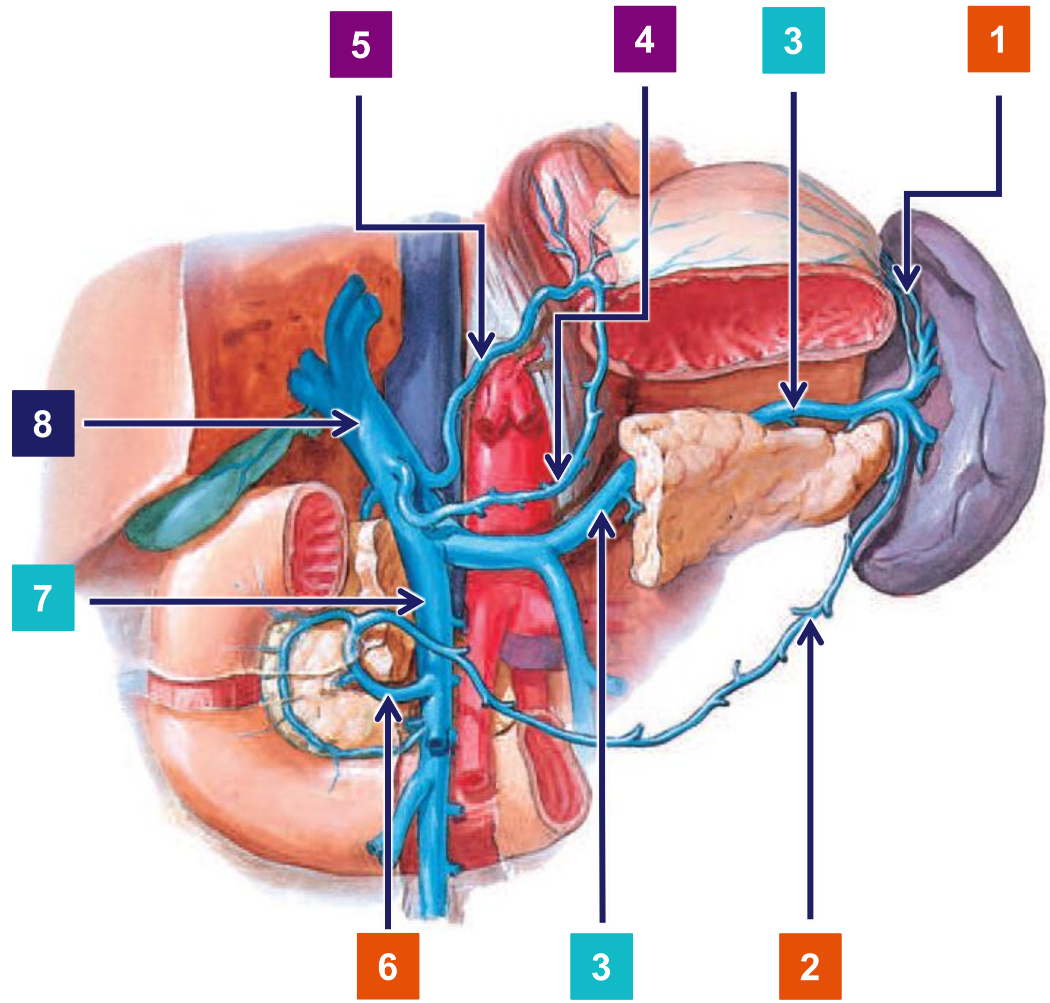
Gastro-omental = Gastroepiploic

Study tip: Draw your own flow diagram

Stomach – Venous Drainage



1. Short gastric
2. L. gastro-omental
3. Splenic
4. Right gastric
5. Left gastric
6. R. gastro-omental
7. *Superior mesenteric*
8. **Hepatic portal vein**



Stomach – Blood Supply

Arteries:

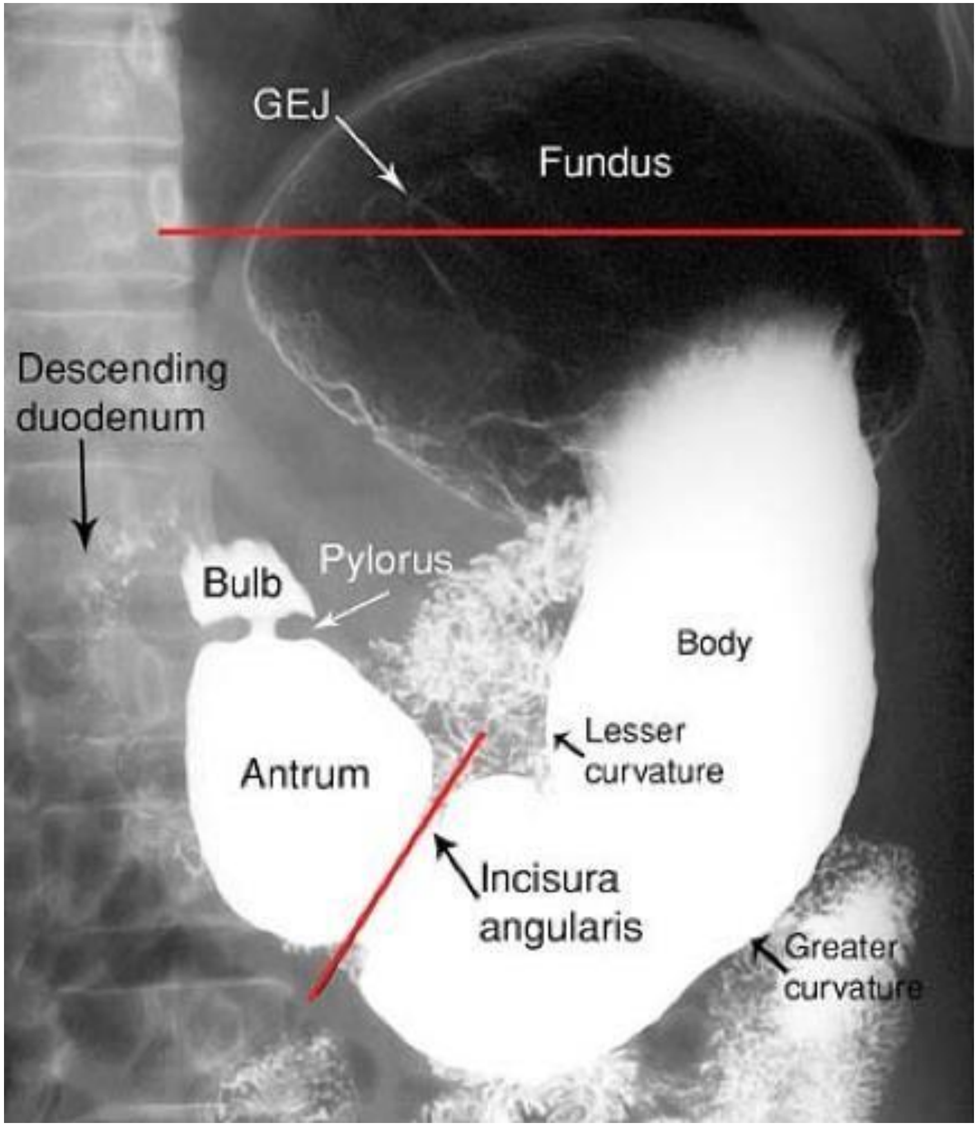
- **Left gastric**: Branch of the celiac trunk. Supplies the lower part of the esophagus and upper left part of the stomach. Lesser ommentum
- **Right gastric**: Arises from the hepatic artery and runs along the lesser curvature of the stomach and lesser ommentum
- **Short gastric**: Arises from the splenic artery. Supplies the fundus.
- **Left gastroepiploic / gastro ommental**: Arises from the splenic artery then passes forward to supply the upper part of the greater curvature of the stomach and greater ommentum
- **Right gastroepiploic / gastro ommentral** : Arises from the gastroduodenal branch of the hepatic artery. Supplies right lower part of stomach along the greater curvature and greater ommentum

Veins:

- Veins all drain into the portal system
- Left and right gastric drain into the portal vein directly
- The short gastrics and left gastroepiploic veins drain into the splenic vein

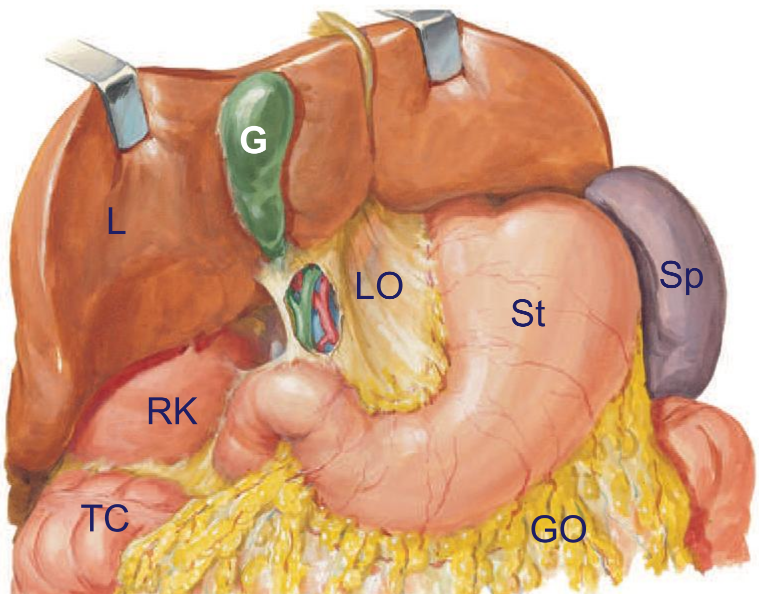
Lymphatic channels from the stomach will first drain to the celiac group of lymph nodes located around the celiac trunk.

Stomach - Imaging



Liver

- Right hypochondrium and epigastric and (left hypochondrial) regions.
- Enclosed in lower portion of rib cage - trauma can be caused by
  - Fractures of lower ribs.
  - Penetrating wounds (thorax or upper abdomen)
  - All can be associated with severe hemorrhage



L – Liver; RK – Right Kidney; TC – Transverse colon; G – Gallbladder; St – Stomach; Sp - Spleen; LO – Lesser Omentum; GO – Greater Omentum

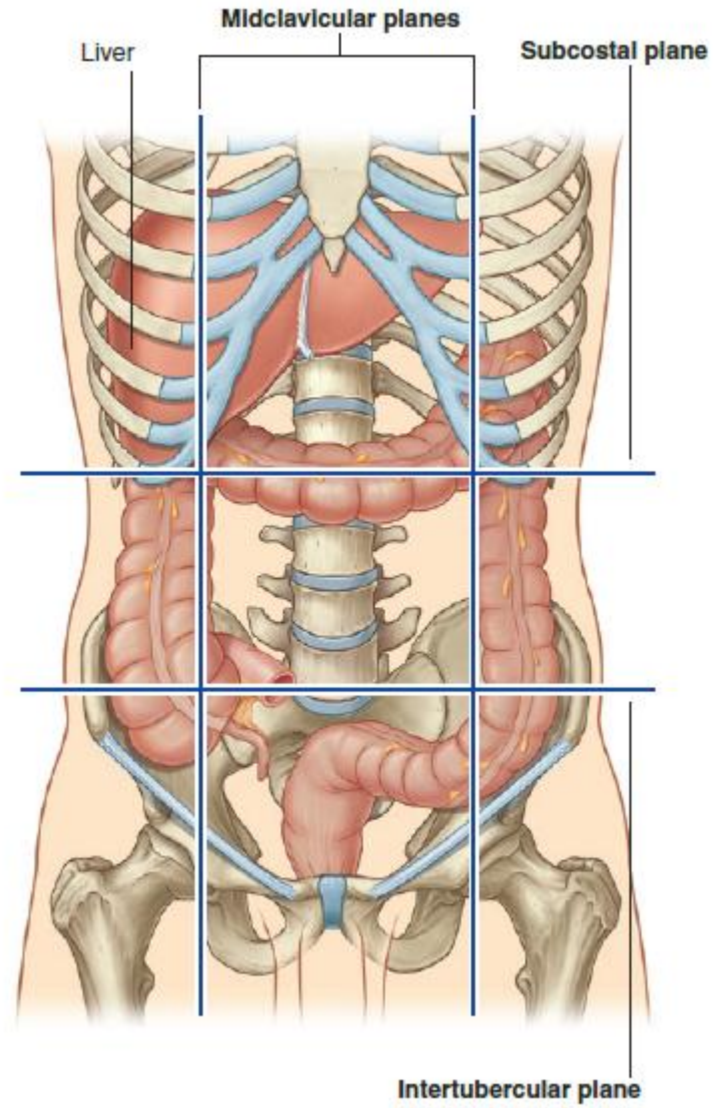
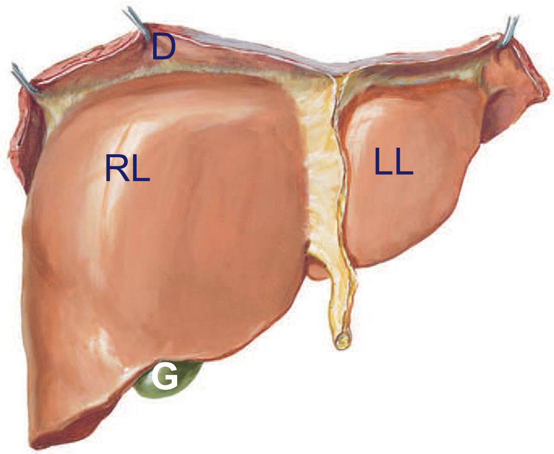
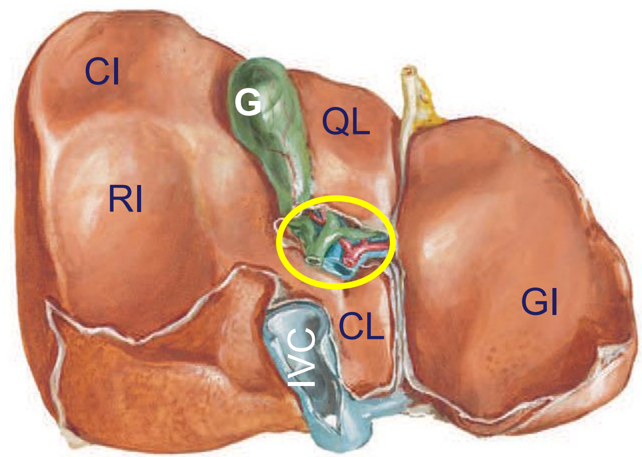


Fig. 4.101 Position of the liver in the abdomen.

Liver Surfaces



Anterior view: RL – Right lobe; LL – Left Lobe; D – Diaphragm; G - Gallbladder



Visceral view: CL – Caudate Lobe; QL Quadrate Lobe; GI – Gastric Impression; RI – Renal Impression; CI – Colic Impression; Circle – Porta Hepatis; G – Gallbladder

<u>**The liver has two surfaces**</u>:

- <u>**Diaphragmatic**</u>: anterior, superior, posterior
- <u>**Visceral surface**</u>: inferior and covered with visceral peritoneum (except the fossa for gallbladder + porta hepatis)

- <u>**Subphrenic space**</u>: between diaphragm and liver
- <u>**Hepatorenal recesses**</u>: [between right kidney & liver] is the gravity dependent part of the peritoneal cavity in supine position

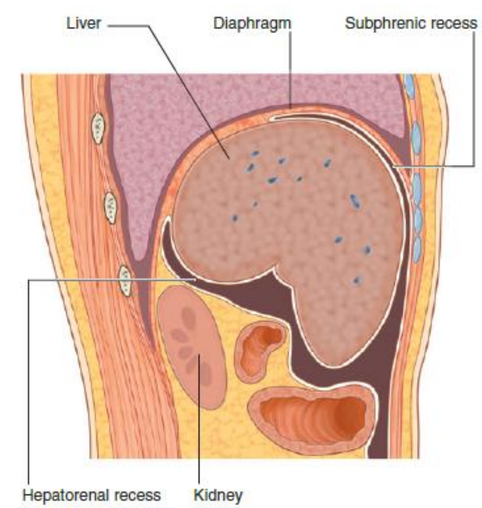
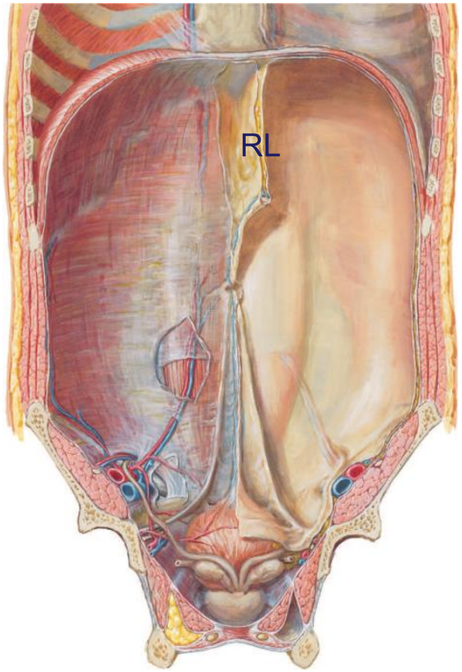
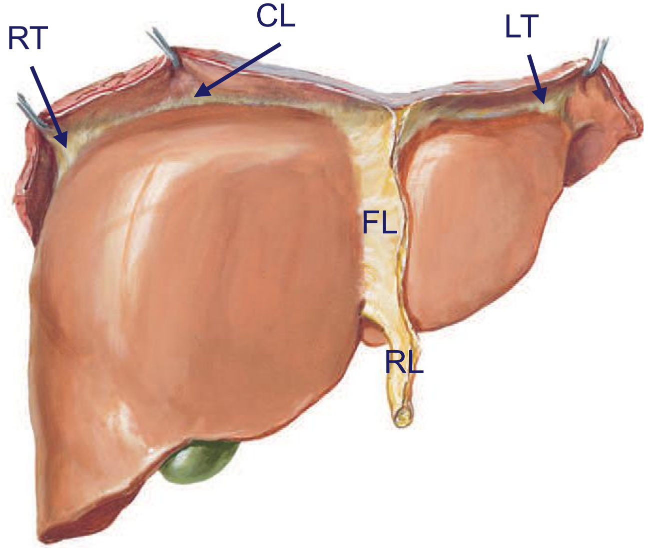


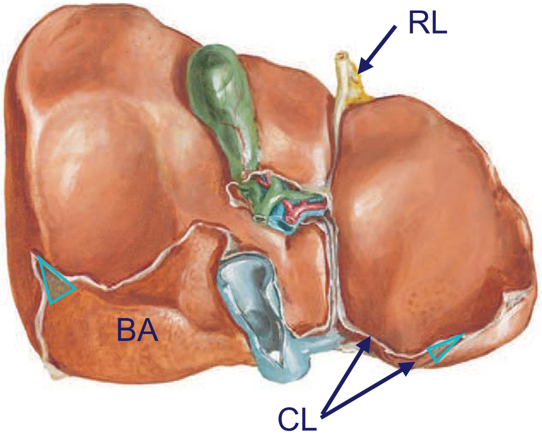
Fig. 4.102 Surfaces of the liver and recesses associated with the liver.

Ligaments of Liver





Anterior view: FL – Falciform Ligament; RL – Round ligament (ligamentum teres hepatis); LT; Left Triangular Ligament; RT – Right Triangular Ligament; CL – Coronary Ligament



Visceral view: BA – Bare Area; RL – Round ligament; Triangle – Triangular ligament; CL – Coronary Ligament

- **Falciform Ligament** – attaches liver to anterior abdominal wall (ventral mesentery)
- **Round Ligament** – posterior free margin of falciform ligament (obliterated umbilical vein)
- **Bare area of liver** – triangular area on diaphragmatic surface devoid of peritoneal covering
- **Coronary Ligaments** – (anterior & posterior) attach the liver to the diaphragm
- **Triangular Ligaments** – (right & left) attach the liver to the diaphragm
- ***Umbilical ligaments** – median, medial (2), lateral (2) Review MSK*

Ligaments of Liver

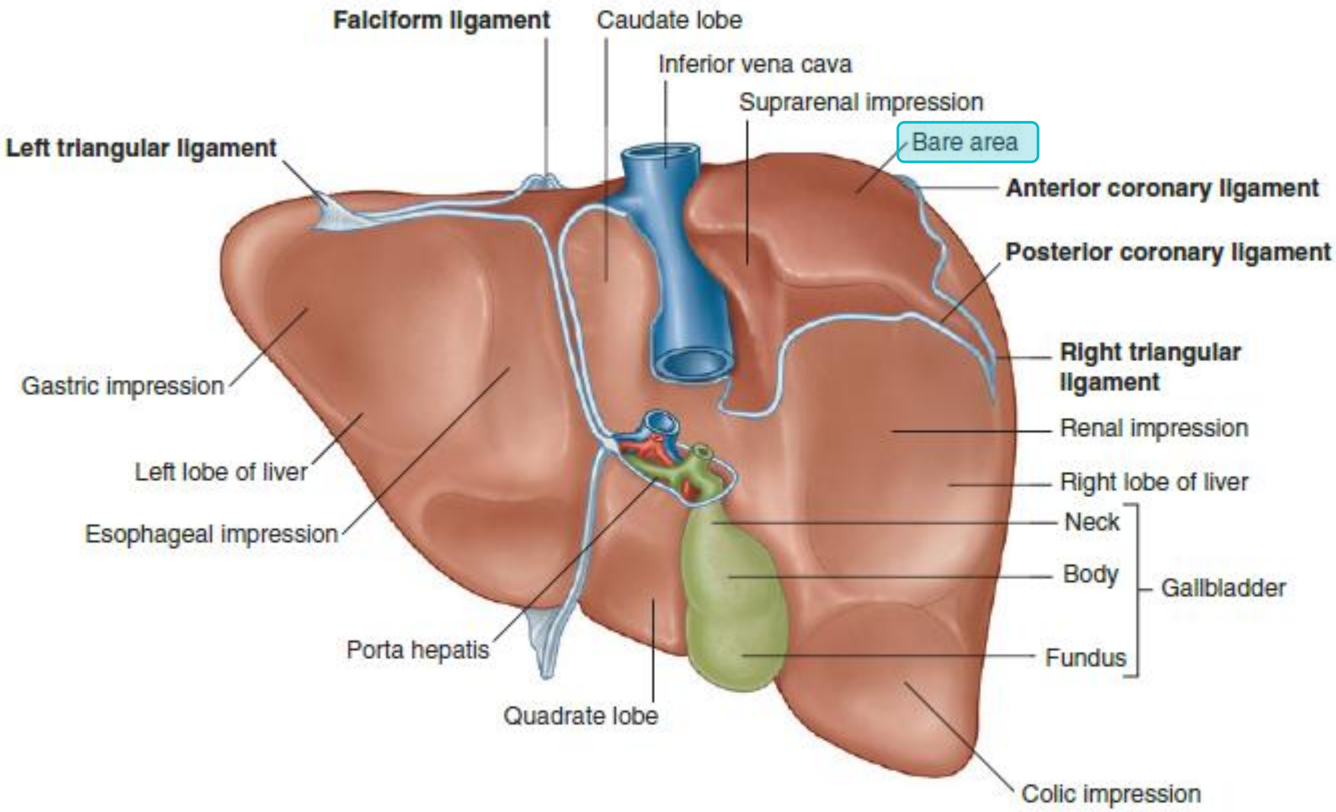
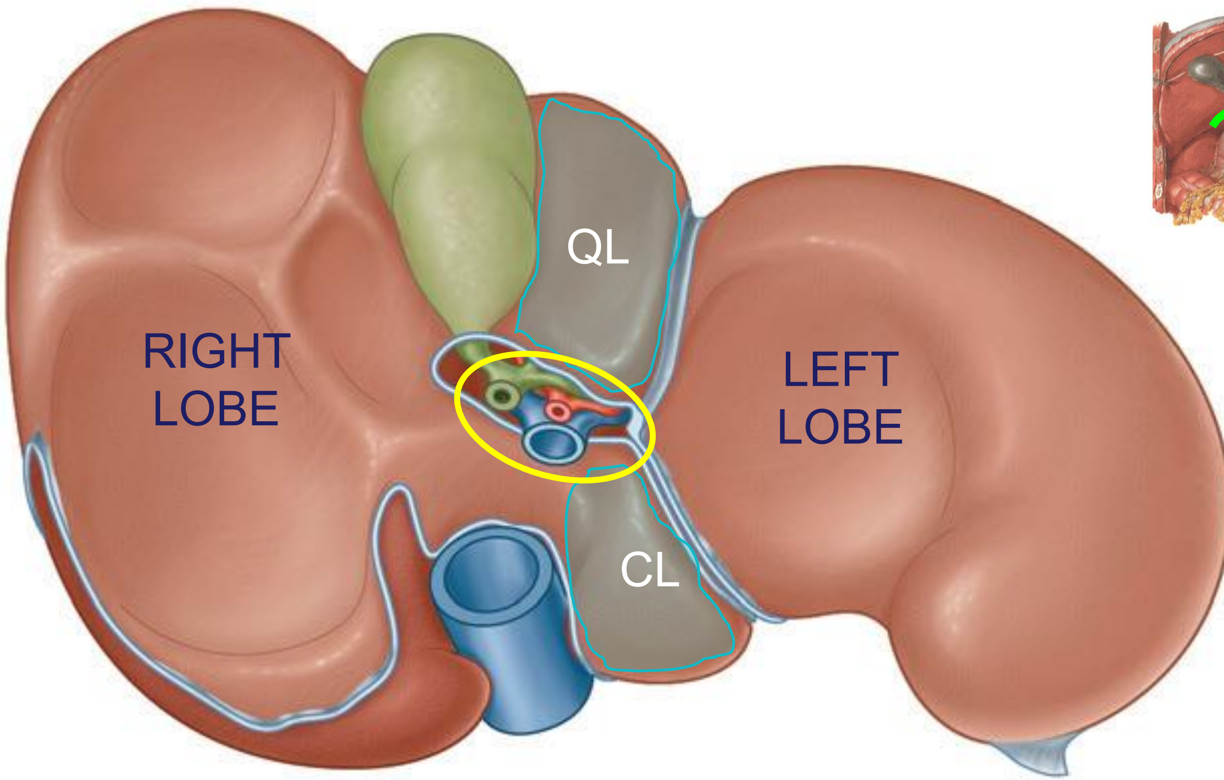


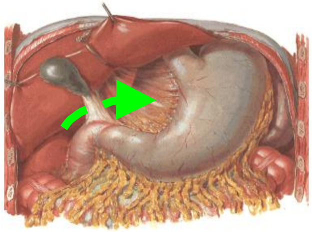
Fig. 4.105 Posterior view of the bare area of the liver and associated ligaments.

- **Falciform Ligament** – attaches liver to anterior abdominal wall (ventral mesentery)
- **Coronary Ligaments** – (anterior & posterior) attach the liver to the diaphragm
- **Triangular Ligaments** – (right & left) attach the liver to the diaphragm
- **Bare area of liver** – triangular area devoid of peritoneal covering, related to the diaphragm

Liver Lobes

- Divided into Left and Right Lobes by falciform ligament
  - Caudate lobe: posterior visceral surface between ligamentum venosum fissure + IVC groove
  - Quadrate lobe: anterior visceral surface between ligamentum teres fissure + gallbladder fossa
  - Anatomically, right lobe larger with caudate & quadrate lobes
- Porta hepatis [circle] – structures enter / leave the liver (except hepatic veins)
  - ENTRY: Hepatic arteries, Portal vein; EXIT: Hepatic bile ducts
  - **Pringle maneuver**: manually compress/clamp hepatoduodenal ligament (which contains vessels of the porta hepatis) anterior to omental foramen to control bleeding to/from liver





Liver Segments

- Liver is divided into 8 segments based on the main branches of right & left hepatic arteries, portal vein & hepatic ducts [hepatic veins → intersegmental]
- Branches of the right & left hepatic arteries, ducts and portal vein do not communicate significantly.
    - [hepatic lobectomy / segmentectomy can be performed]

Summary – Liver Lobes

- **Anatomically,** the liver can be divided into four lobes: right ,left, caudate and quadrate. In this classification the quadrate and caudate lobes are considered part of the right liver.
- **Functionally,** it can be divided into independent right and left livers i.e., portal lobes. This functional division is produced by the sagittal plane passing through the gall bladder fossa and the fossa of the IVC on the visceral surface of the liver, and an imaginary line over the diaphragmatic surface that runs from the surface of the gall bladder to the IVC. The left liver now includes anatomic quadrate and caudate lobes. This functional division results in the liver having a slightly larger right lobe. These "portal lobes" will have their own blood supply from the hepatic artery, portal veins, their own venous (hepatic veins) and biliary drainage.
- The (2) portal lobes of the liver are further divided into 8 segments. The segmentation is based on the principal branches of the right and left hepatic arteries, hepatic portal veins and hepatic ducts . Each segment is supplied by the corresponding branch of the right or left hepatic artery, portal vein and drained by a branch of the left or right hepatic duct.

Moore, K. L., Agur, A. M., & Dalley, A. F. (2002). *Essential clinical anatomy* (Vol. 469). Philadelphia: Lippincott Williams & Wilkins.

Liver – Blood Supply

Arterial supply – hepatic arteries

- Hepatic artery (proper) (2) [30%]
- Right (3) and Left (4) hepatic a.
- Portal vein (5) [70%] to the liver

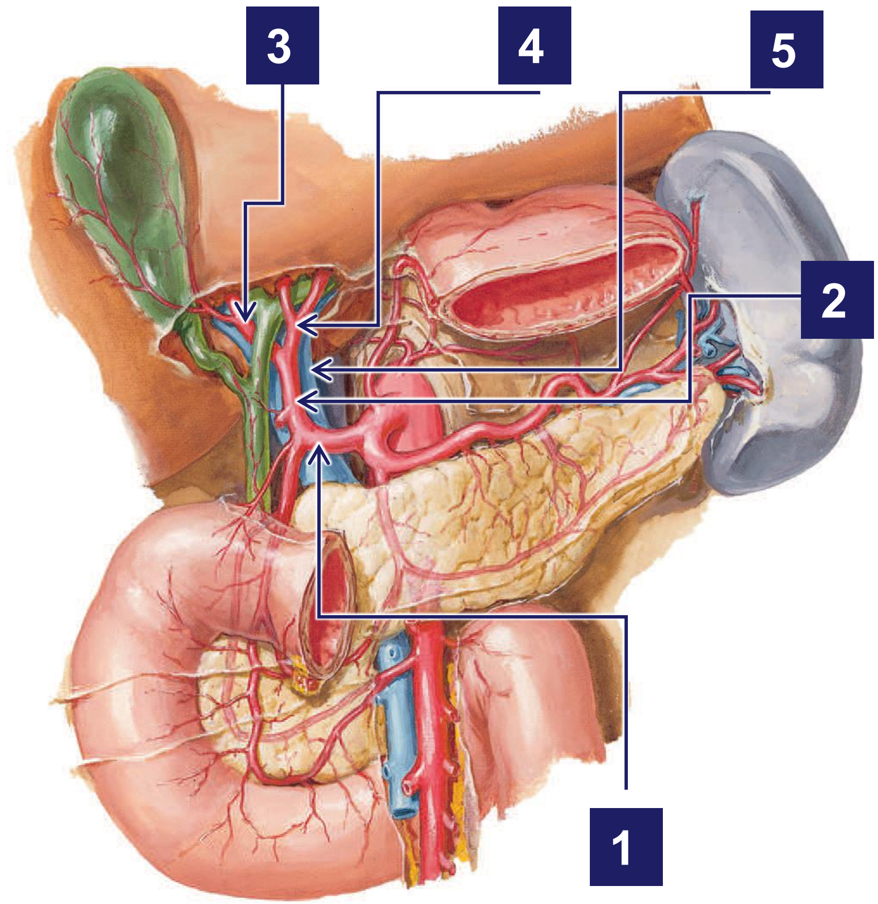
Veins – hepatic & portal

- Hepatic veins open into IVC

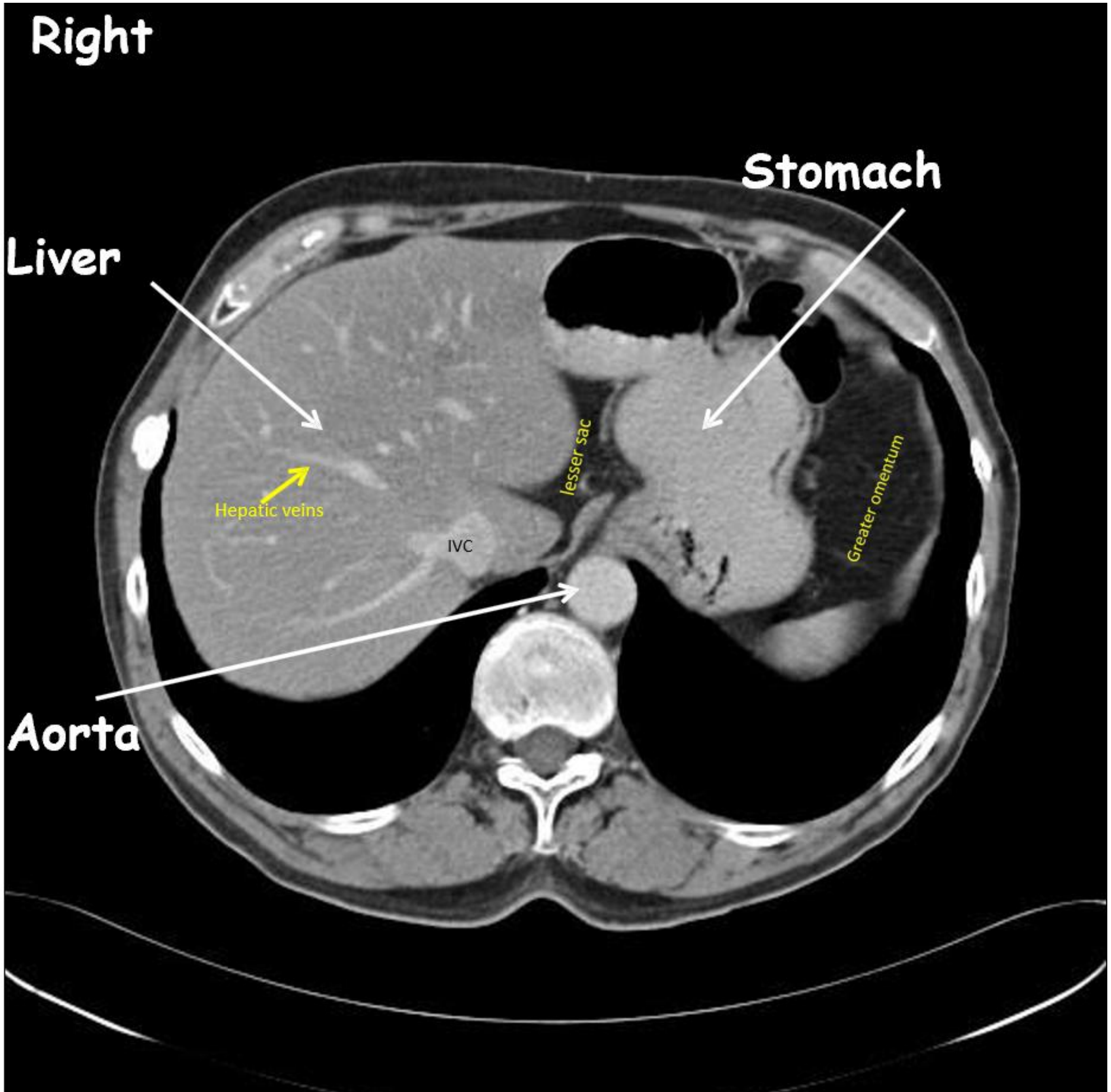
Liver produces large amount of lymph

- → mainly to celiac nodes

*1. Common hepatic artery



Liver Imaging



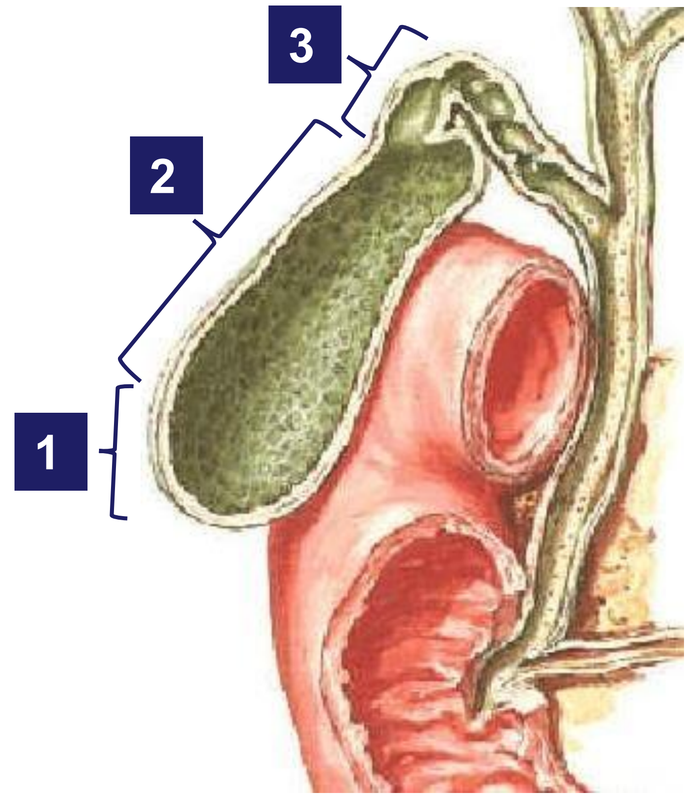
Gallbladder

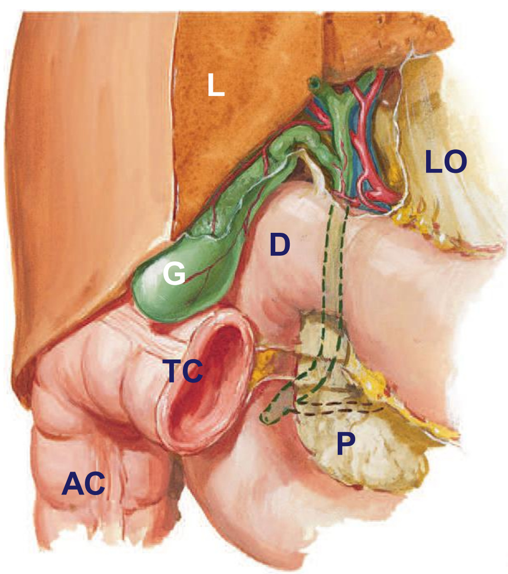
Pear shaped, in its fossa on the inferior surface of liver

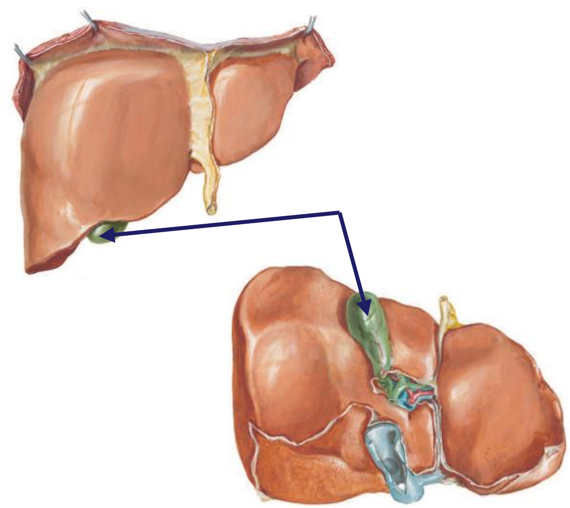
<u>Parts</u>

1. **Fundus:** rounded end [projects beyond the inferior margin of liver → located at the tip of the right 9th costal cartilage mid-clavicular line],
2. **Body:** major part, against TC and D
3. **Neck:** narrow part → leads to cystic duct

Receives bile from the liver (concentrates & stores it)

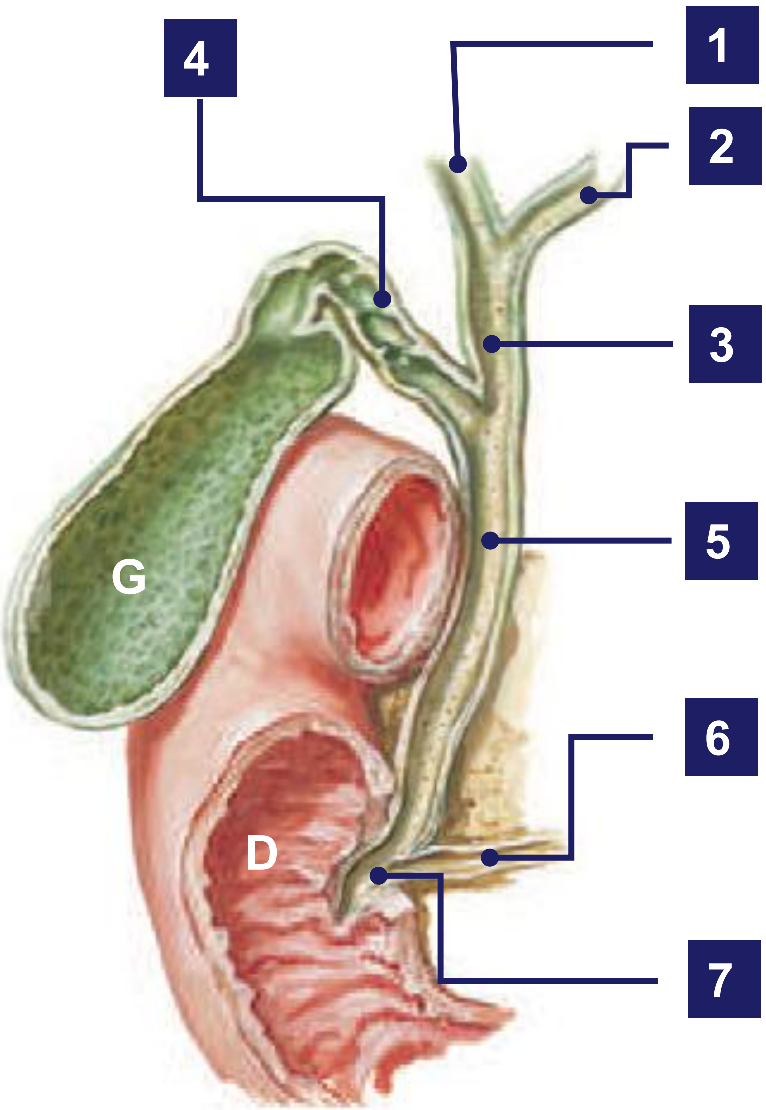






L – Liver; G – Gallbladder; TC – transverse colon; AC – Ascending colon; D – Duodenum; P – Pancreas; LO – Lesser Omentum

Biliary Apparatus



- Right (1) & left (2) hepatic ducts join to form the common hepatic duct (3);
- Joined by the cystic duct (4) to form the [common] bile duct [BD] (5)
- BD passes posterior to the 1st part of duodenum
- Joins the pancreatic duct (6) to open into the major duodenal papilla (7)

1. Right hepatic duct
2. Left hepatic duct
3. Common hepatic duct
4. Cystic duct
5. [Common] Bile duct
6. Main pancreatic duct
7. Hepato-pancreatic Ampulla of Vater

G – Gallbladder; D - Duodenum

Biliary Apparatus – Blood Supply

Arterial
- Cystic artery [1]
  - from R hepatic a. [2]

Venous
- Cystic veins (portal v.)

Lymphatics
- Celiac trunk nodes

Triangle of Calot

Boundaries:
- Superior – Liver
- Lateral – Cystic duct
- Medial – Common hepatic duct

Contents: Cystic artery